



SEMANTIC SEO: THE SHIFT FROM STRINGS TO THINGS

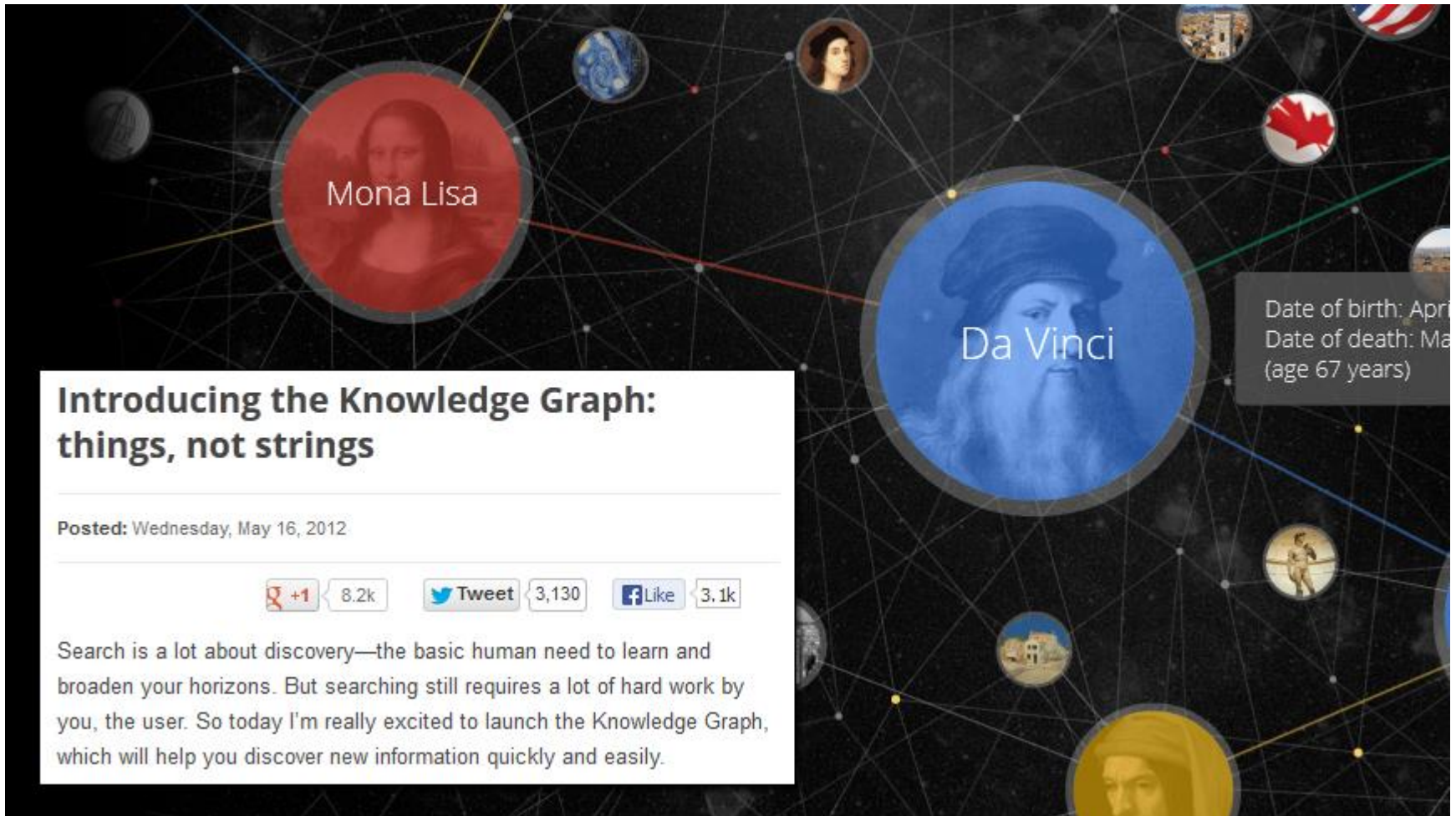


InfoMine
Mining Intelligence & Technology

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InfoMine, Inc.
@Aaranged
#SMX #21C
October 2, 2013

“Things, not strings”

The phrase coined by Google to highlight the shift to real-life things in search



**Introducing the Knowledge Graph:
things, not strings**

Posted: Wednesday, May 16, 2012

 +1 8.2k  Tweet 3,130  Like 3.1k

Search is a lot about discovery—the basic human need to learn and broaden your horizons. But searching still requires a lot of hard work by you, the user. So today I'm really excited to launch the Knowledge Graph, which will help you discover new information quickly and easily.

Da Vinci
Date of birth: April 15, 1453
Date of death: May 2, 1519
(age 67 years)

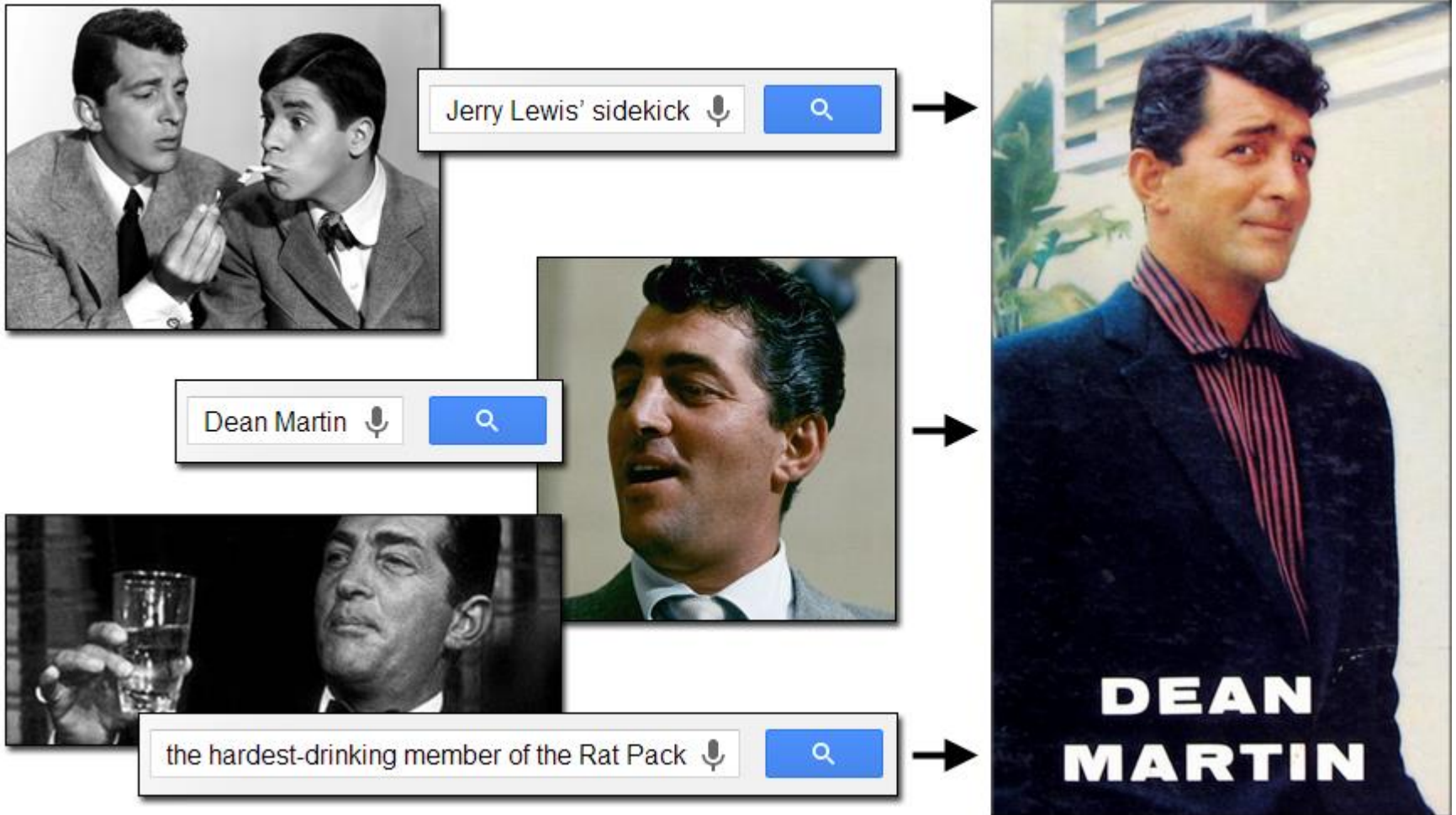
Semantic search and the SEO strategy gap

So far most semantic SEO work has been carried out in a strategic void



Different keywords, same entity

Keywords are not a reliable way of designating real-life things



Same keyword, different entities

Keywords are not a reliable way of designating real-life things

Photo: agaw.dilim / Flickr

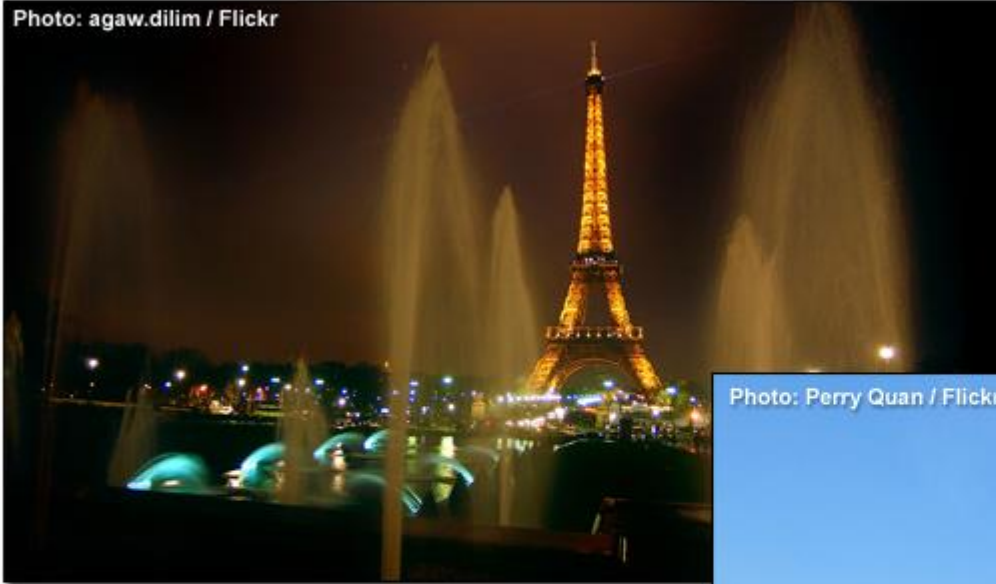


Paris

Same keyword, different entities

Keywords are not a reliable way of designating real-life things

Photo: agaw.dilim / Flickr



Paris (France)

Paris (Ontario, Canada)

Photo: Perry Quan / Flickr



Unique identifiers

For a given domain an identifier, not keywords, represents the thing itself

<http://en.wikipedia.org/wiki/Eggplant>



<http://www.imdb.com/title/tt0063350/>



Triples: subject > predicate > object

Semantic applications use a standard model for describing resources

subject → predicate → object



Dean Martin → has height → 5'10"



Acme 8 GB USB drive → has price → \$9.99



United Flight U323 → has departure time → 16:10

BONUS FACT FOR KEENERS: this is the model used by the **Resource Description Framework (RDF)**

Triples in action

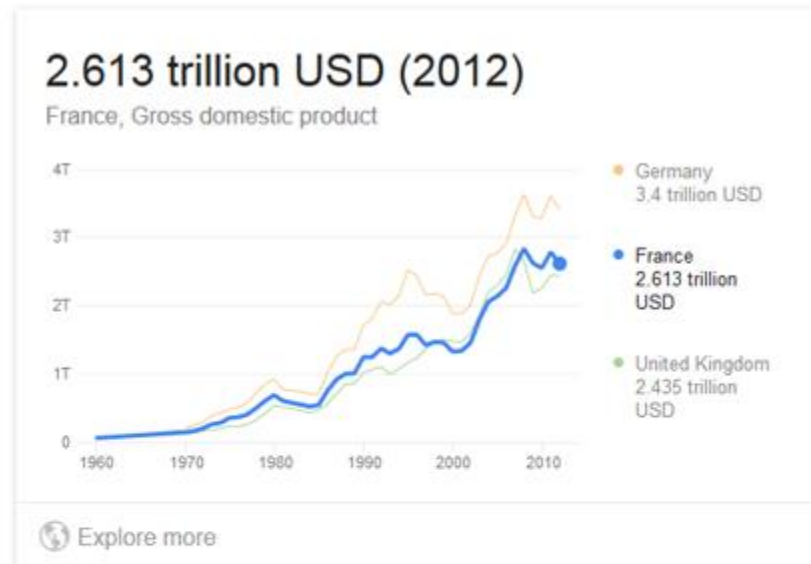
schema.org properties express relationships between a subject and an object

```
<div itemscope itemtype="http://schema.org/Product">  
  <h1 itemprop="name">Acme 8 GB USB Drive</h1>  
</div>
```

product → has name → Acme 8 GB USB Drive

The GDP of France ... and a lot more

From structured sources, and/or sources Google has parsed and structured



Sources include: World Bank

[France GDP Growth Rate | Actual Value | Historical Data | Forecast](#)

www.tradingeconomics.com/france/gdp-growth

Aug 14, 2013 - France GDP Growth Rate averaged 0.46 Percent from 1978 ...

[France GDP | Actual Value | Historical Data | Forecast | Calendar](#)

www.tradingeconomics.com/france/gdp

The Gross Domestic Product (GDP) in France was worth 2612.90 billion US dollars in 2012. The GDP value of France represents 4.21 percent of the world ...

[France GDP Data & Country Report | Global Finance](#)

www.gfmag.com > Archives > June 2012

France GDP statistics and forecasts; France economic, financial and trade information; best banks and safest banks in France; doing business in France.

France
Country

France, officially the French Republic, is a unitary semi-presidential republic in Western Europe, with several overseas regions and territories.
[Wikipedia](#)

Related statistics

Population	65.7 million (2012)
Unemployment rate	10.9% (May 2013)
GDP growth rate	0.0% annual change (2012)

Gross domestic product elsewhere

United States of America	15.68 trillion USD (2012)
Italy	2.013 trillion USD (2012)
Spain	1.349 trillion USD (2012)

Sources include: World Bank, Eurostat

[Feedback / More info](#)

Related queries: strings vs. things

Semantic search allows Google to determine the meaning behind queries

Strings

- gdp of britain
- what's the gdp of the uk
- britain gross domestic product
- united kingdom gdp
- r-u produit interieur brut
- gdp data u.k.
- gdp of gr. britain
- united kingdom gross domestic product

Total queries: ?

Things



Total queries: X

Traditional definition of “SEO”

For most of its history SEO has been focused on keywords

SEO:

Optimizing for
keywords

blender

Acme

John Doe

Toward a definition of “semantic SEO”

Entities are a necessary component – but only a component – of semantic SEO

Semantic SEO:

Optimizing for
entities
only in part

<http://en.wikipedia.org/wiki/Blender>



http://en.wikipedia.org/wiki/Acme_Inc.

ACME[®]
SUPERIOR KITCHEN PRODUCTS

<https://plus.google.com/106943062990152739506>

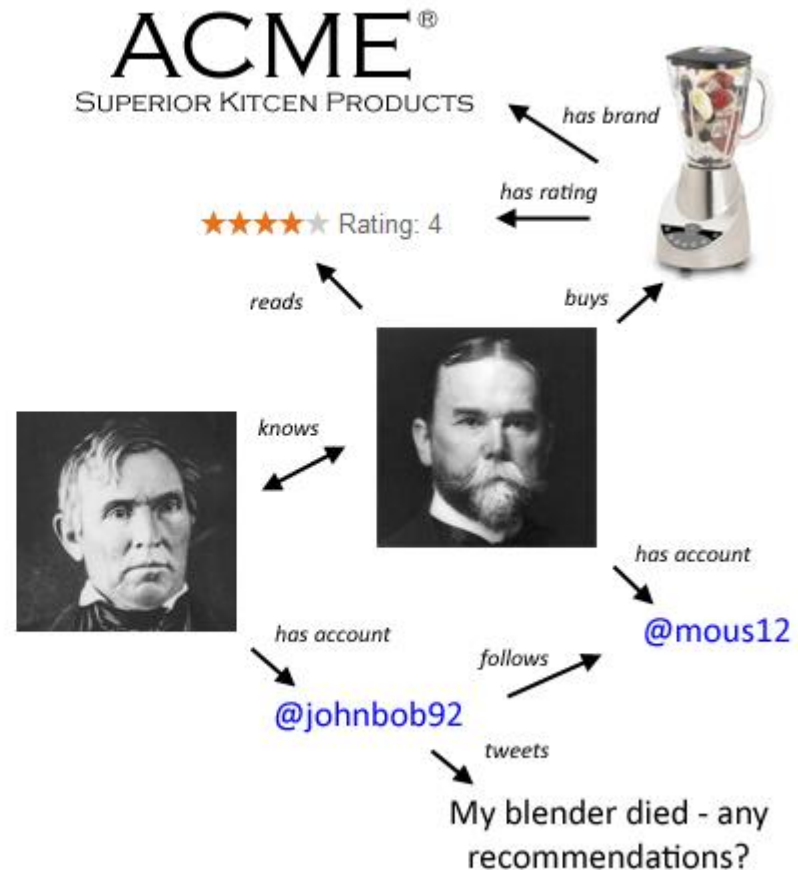


“Semantic SEO”

A shift in focus from matching strings to the relationships between things

Semantic SEO:

Optimizing for
relationships



STRATEGY: identify and disambiguate entities

Extraction utilities and APIs can aid in identifying named and topical entities

POMEGRANATE JUICE MADE EASY

MAY 15, 2013 | POSTED BY ADMIN

Have you watched any of the **Pomegranate Juice** videos explaining the many different ways to make **Pomegranate Juice**? Kudos goes out to all health minded people with the tenacity and patience for all the messy cleanup but let's be realistic, just separating the seeds.

Meanwhile, the **Pomegranate Juice** Concentrate is available at home that can save you time.

The formula for doing so goes something like this:

1. Call our toll free number 1-877-591-3100. Visit our **Order** page, add your item(s) to your cart.
2. Within a week, your order will arrive. Place it in the freezer (our **Pomegranate Juice Concentrate** is frozen).
3. Pour 1 ounce (2 tablespoons) of **FruitFast** Pomegranate Juice Concentrate into a glass, mix well and add ice cubes. I prefer using a blender.
4. Bottoms up!

It's that simple, no deseeding, no squeezing seeds. Just **Concentrate** that's grown in California, and you're done.



Named Entity Extraction

Identify people, companies, organizations, cities, geographic features, and other named entities within your content. [Learn more.](#)

Note: All entities are annotated above in **blue**. Click any rows below to highlight that entity in **red**.

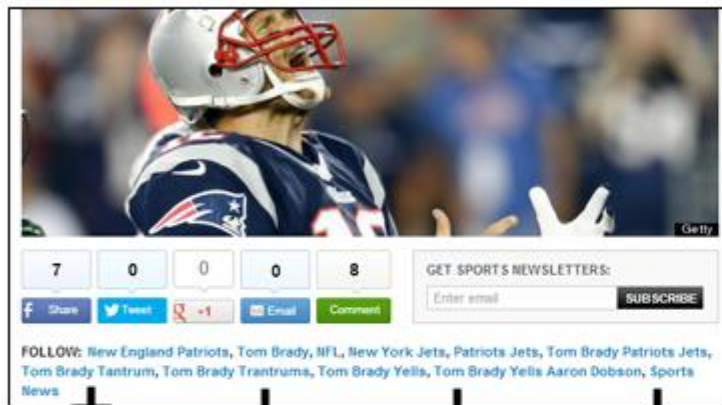
Entity	Relevance	Sentiment	Type	Subtypes	Linked Data
FruitFast	0.792216	positive	Company		
California	0.682942	neutral	StateOrCounty	Location, PoliticalDistrict, AdministrativeDivision, Appellation, GovernmentalJurisdiction, USState, WineRegion	Website Lat 35.46, Lon: -119.36 DBpedia Yago OpenCyc

- Entities
- Keywords
- Concepts
- Sentiment
- Text
- Author
- Language
- Title
- Relations
- Quotations
- Category
- Feeds
- Microformats

Entities: only as many identifiers as necessary

Entity cannibalization is the new keyword cannibalization: just say no

Strings



Things



STRATEGY: identify items and properties

Structuring your content will help when it comes to structuring it for SEs

Strings

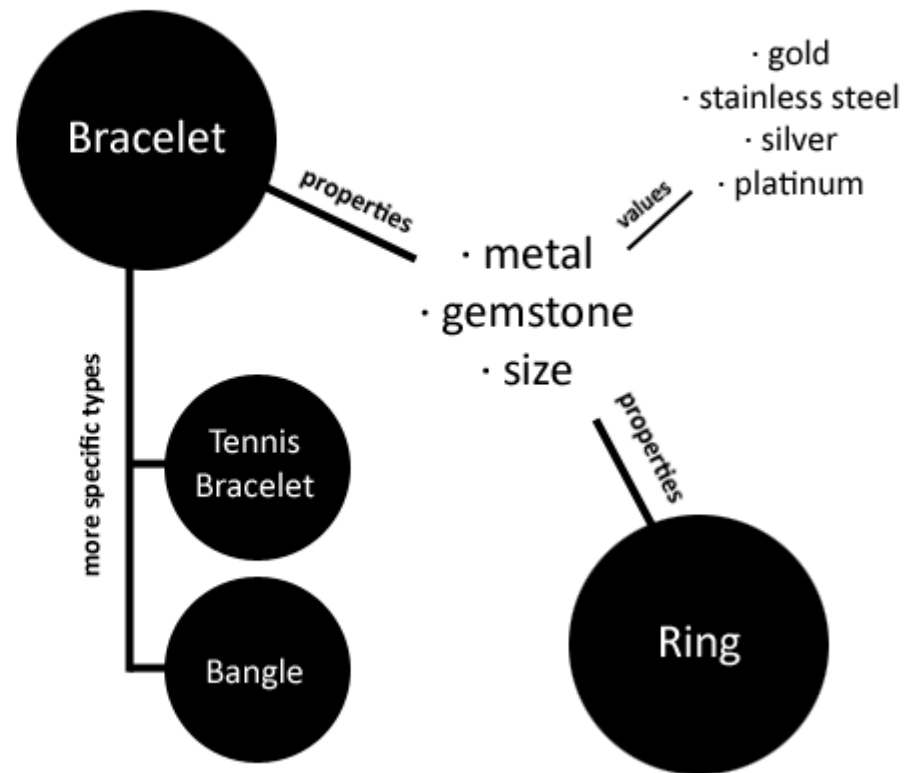
BRACELETS

- gold bracelets
- silver bracelet
- diamond bracelet
- gemstone bracelet

BANGLES

- gold bangle
- amethyst stack bangle
- bangle jewelry
- resin bangle

Things



STRATEGY: declare your data

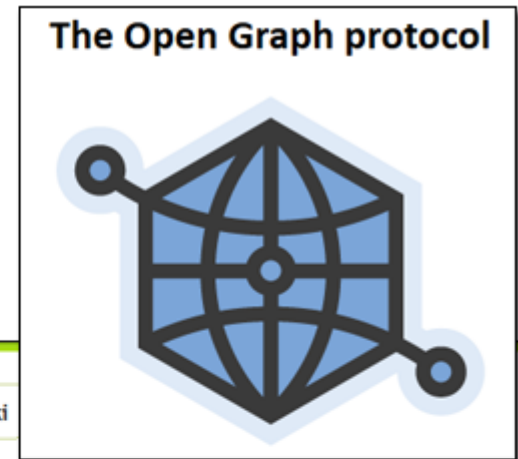
Provide your data in a structured format for machine consumption

schema.org

Home Schemas Documentation

Thing > CreativeWork > Recipe
A recipe.

Property	Expected Type	Description
Properties from Thing		
additionalType	URL	An additional type for the more specific types from microdata syntax. This is and a class that the thing to use the native RDFa sy multiple types. Schema.o understanding of extra t externally.
description	Text	A short description of the
image	URL	URL of an image of the i
name	Text	The name of the item.



microformats blog wiki

search Go Search

[View source](#) [History](#)

h-product

[Tantek Çelik](#) (Editor)

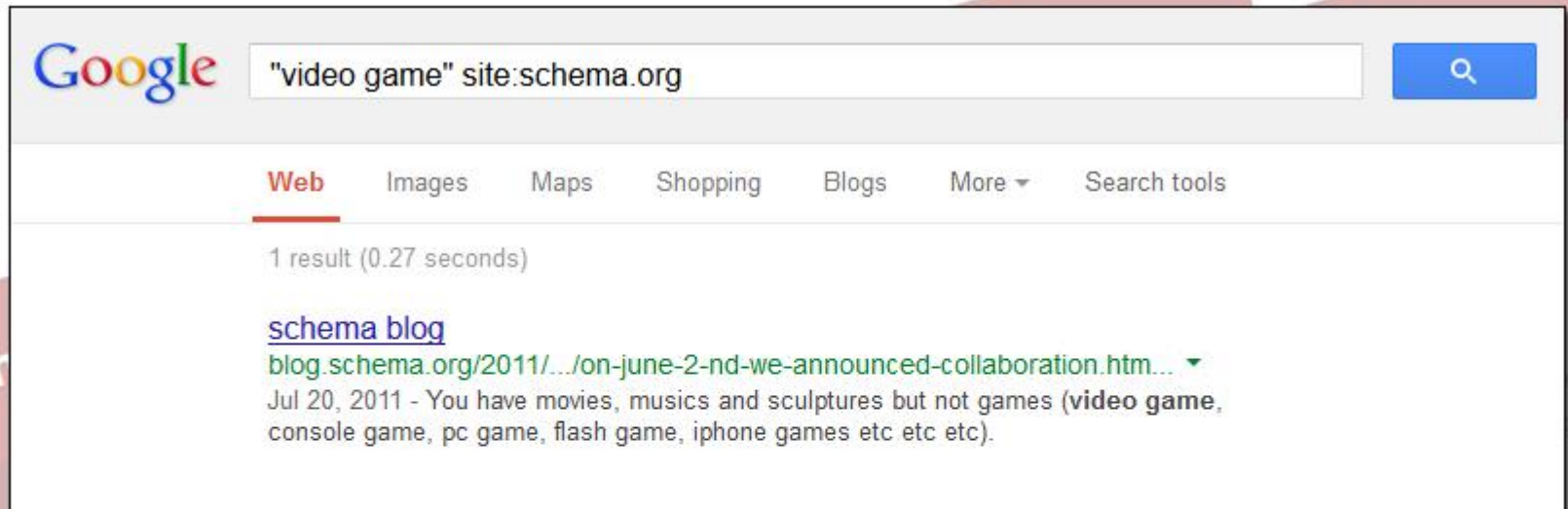
h-product is a simple, open format for publishing product data on the web. h-product is one of several open [microformat](#) draft standards suitable for embedding data in HTML/HTML5.

h-product is the [microformats2](#) update to [hProduct](#).

Per [CC0](#), to the extent possible under law, the editors have waived all copyright and related or neighboring rights to this work. In addition, as of 2013-09-13, the editors have made this specification available under the [Open Web Foundation Agreement Version 1.0](#).

Missing in action ... but not necessarily forever

If a resource is absent in schema.org you can add it as an extension



Extension Mechanism

Many applications, including search, can make use of structured data only. However, the variety and richness of structured data covering everything on the web is not completely covered by schema.org. Therefore, schema.org has created an extension mechanism that lets webmasters and developers extend our existing schema.org vocabulary. Extensions are not at all tied to those on schema.org, and you should use them as you see fit. If the schema gains adoption and proves useful to a large number of people, it may be added to the schema.org vocabulary.

No snippet? No problem!

Google still wants your data

The screenshot displays the Google Data Highlighter interface. On the left, a list of properties for 'TVSeries' is shown, including actor, actors, director, endDate, episode, episodes, musicBy, numberOfEpisodes, producer, productionCompany, season, seasons, startDate, and trailer. The 'musicBy' property is highlighted in yellow. In the center, a video player shows a man speaking, with a play button overlay. To the right, a list of properties for 'Restaurant' is shown, including Name, Address, Telephone, Opening hours, Cuisine, Image, URL, Reservation URL, Menu URL, Average rating, Rating, Votes, Review, Reviewer, Review rating, and Review date. The 'Cuisine' property is highlighted in yellow. A black box with the text 'Highlight data on the page to the left' is positioned over the 'Cuisine' property. A black arrow points from this box to the 'musicBy' property in the TVSeries list.

Properties from **TVSeries**

<u>actor</u>	<u>Person</u>	A cast member of the movie, TV series, season, or episode, or video.
<u>actors</u>	<u>Person</u>	A cast member of the movie, TV series, season, or episode, or video. (legacy spelling; see actor)
<u>director</u>	<u>Person</u>	The director of the movie, TV episode, or series.
<u>endDate</u>	<u>Date</u>	The end date and time of the event (in <u>ISO 8601 date format</u>).
<u>episode</u>	<u>TVEpisode</u>	An episode of a TV series or season.
<u>episodes</u>	<u>TVEpisode</u>	The episode of a TV series or season (legacy spelling; see singular form, episode).
<u>musicBy</u>	<u>MusicGroup or Person</u>	The composer of the movie or TV soundtrack.
<u>numberOfEpisodes</u>	<u>Number</u>	The number of episodes in this season or series.
<u>producer</u>	<u>Person</u>	The producer of the movie, TV series, season, or episode, or video.
<u>productionCompany</u>	<u>Organization</u>	The production company or studio that made the movie, TV series, season, or episode.
<u>season</u>	<u>TVSeason</u>	A season of a TV series.

Overview About

Data Highlighter. Show off your site's data in search.

Introduction to Data Highlighter

Does your site contain structured data like event listings or review ratings?

Now there's a new alternative to webmaster markup for helping Google understand your site's data. With Data Highlighter, you simply "tag" each data field with your mouse.

Then Google can present your data more attractively, and in new ways, in search results.

[Start Highlighting](#)

Restaurant

Name: required

Address: required

Telephone: required

Opening hours:

Cuisine:

Image: Highlight data on the page to the left

URL:

Reservation URL:

Menu URL:

Average rating:

Rating:

Votes:

Review

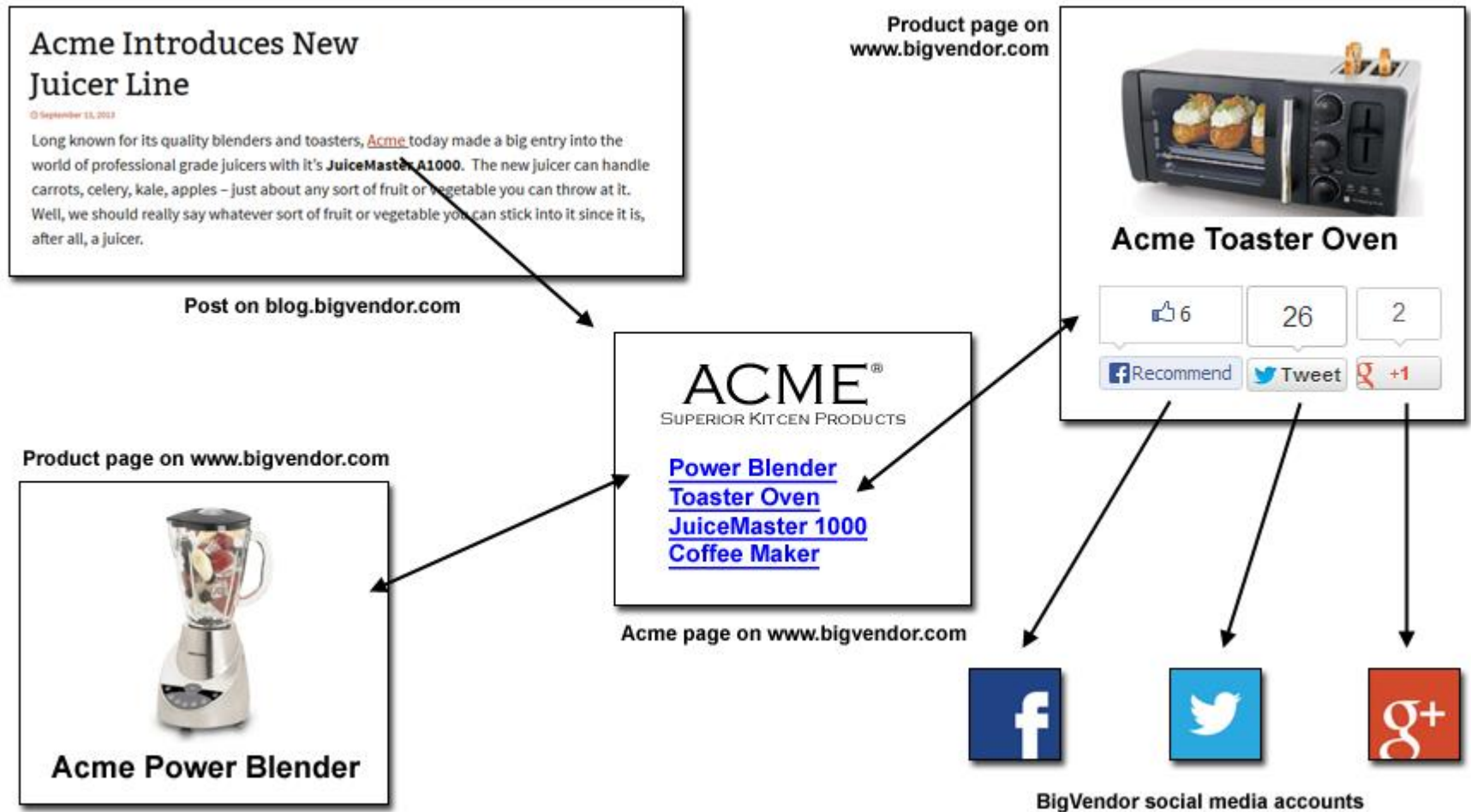
Reviewer:

Review rating:

Review date:

STRATEGY: map and declare your relationships

Hyperlink relevant entities and resources, preferably with structured data



STRATEGY: build trust in your data

Trust by verification: Google+, Google's ~~social~~ verified identity network

The screenshot shows a Google+ post from the Getty Museum. The post header includes the name 'Getty Museum', a verified badge, and a 'Follow' button with '+1,419' followers. The main content of the post is a link to 'EduMine - Professional Development and Training for Mining' with the URL 'www.edumine.com/'. Below the link, it says 'by EduMine · More by EduMine ·' and a placeholder text: 'The excerpt from the page will show up here. The reason we can't show text from your webpage is because the text depends on the query the user types.' Two verification overlays are present. The first overlay, titled 'Publisher', states 'Publisher markup is verified for this page.' and shows a 'Linked Google+ page:' as 'https://plus.g...'. The second overlay, titled 'Authorship rel=author Markup', states 'rel=author markup has successfully established authorship for this webpage.' and provides details: 'First rel=author link from webpage: http://daggle.com/author/danny-sullivan', 'Link (direct or indirect) from website to Google+ profile: Yes', and 'Public contributor-to link from Google+ profile to daggle.com: Yes'.

Getty Museum

Follow +1,419

getty.edu

EduMine - Professional Development and Training for Mining
www.edumine.com/
by EduMine · More by EduMine ·
The excerpt from the page will show up here. The reason we can't show text from your webpage is because the text depends on the query the user types.

Publisher

Publisher markup is verified for this page.

Linked Google+ page: <https://plus.g...>

Skip The Drink Ticket, Southwest Airlines: Give Me A Wifi Pass ...
daggle.com/southwest-wifi-3388
by Danny Sullivan · More by Danny Sullivan
The excerpt from the page will show up here. The reason we can't show text from your webpage is because the text depends on the query the user types.

Authorship rel=author Markup

rel=author markup has successfully established authorship for this webpage.

First rel=author link from webpage: <http://daggle.com/author/danny-sullivan>
Link (direct or indirect) from website to Google+ profile: Yes
Public contributor-to link from Google+ profile to daggle.com: Yes

1200 Getty Center Dr Los Angeles, California 90049
(310) 440-7300 · getty.edu

Art Museum, Non-Profit Organization · Today 10:00

The J. Paul Getty Museum at the Getty Center featu
8th through the 21st century, against a backdrop of
gardens and views of Los Angeles.

Visit this place

Avail yourself of *all* verification methods

Verify identities and data with search engines and social networks

Verify and claim your website URL

Before you can [upload and publish product data](#) to Google Merchant Center, you must verify and claim your website.

The Website URL setting in your Google Merchant Center account is important because it verifies you have the right to upload product listings for the website specified in your account. The links provided in your data feed must match the URL, including the top level domain (ex: .com, .net...) registered in your Google Merchant Center account. Product listings submitted for any URL other than the one registered in your Google Merchant Center account will not process successfully.

1. Determine the exact website URL to verify and claim

To ensure all your uploaded listings are accepted by Google, you'll need to determine the right website URL to verify.

For simple stores, the best URL is the one that appears in your data feed. For example, if your store is at <http://example.com>, and your data feed includes those in your URL, e.g.

Validate your rich pins and apply to get them on Pinterest

1. Make sure you've read the [rich pin documentation](#).
2. Validate your rich pins below by entering a URL below. This could be a product page (a [metatags \(preview\)](#), an oEmbed endpoint ([preview](#)) or a Shopify link ([preview](#)).
3. Preview your pins to make sure they're correct.
4. Once your rich pins are validated, apply to get them approved.

<http://seocooking.com/an-authentic-paella-recipe-semantic-markup-wont-ha>

Validate

✓ Your pin's been validated! [Apply now.](#)

Try Cards

Validate & Apply

Standard Tags

twitter:card

summary

Choose Card...

twitter:site

@seocooking

twitter:title

Verifying methods

twitter:description

By verifying your data, you'll increase the de

twitter:creator

@aaranged

Building trust with data consistency

The search engines will trust your data more if it's consistent across domains

HTML

```
<head>
  <title>Acme Toaster Oven</title>
</head>
```

Open Graph

```
<meta property="og:title" content="Acme Toaster Oven">
```

Google Merchant Center Feed

```
<entry xmlns:scp="http://schemas.google.com/structuredcontent/2009/products">
  <title>Acme Toaster Oven</title>
</entry>
```

Twitter Card

```
<meta name="twitter:title" content="Acme Toaster Oven">
```

RSS

```
<item>
  <title>Acme Toaster Oven</title>
</item>
```

schema.org

```
<body itemscope itemtype="http://schema.org/WebPage">
  <h1 itemprop="name">Acme Toaster Oven</h1>
</body>
```

Tweet Button

```
<a href="https://twitter.com/share" class="twitter-share-button"
data-text="Acme Toaster Oven">Tweet</a>
```

Building trust in ecommerce environments

Products should be uniquely identified, and ecommerce data kept consistent

```
<item>
  <title>Acme Toaster
  Oven</title>

  <link>http://www.acmestuffinc.com
  /toaster3982</link>
  <description>An amazing
  toaster that's also an
  oven!</description>
  <g:id>toaster_3982</g:id>
  <g:condition>new</g:condition>
  >
  <g:price>155.00 USD</g:price>
  <g:availability>in
  stock</g:availability>

  <g:image_link>http://www.acmestuf
  finc.com/images/toaster-oven.jpg<
  /g:image_link>
  <g:shipping>
    <g:country>US</g:country>

  <g:service>Standard</g:service>
    <g:price>14.95
  USD</g:price>
  </g:shipping>
```



Acme Toaster Oven



Rated 4/5 based on 1 customer review

\$155.00 In stock

Product description: an amazing toaster that's also an oven!

Customer reviews:

Amazed - by Phillip, April 1, 2011
4/5 stars

It turned my bread into toast!

Item 2

type: <http://schema.org/offer>

property:

price: \$155.00

availability: <http://schema.org/InStock>

Item 3

type: <http://schema.org/review>

property:

name: Amazed

author: Phillip

datepublished: 2011-04-01

reviewrating: *Item 4*

description: It turned my bread into toast!

Outcomes: increased search visibility

From authorship to answer boxes rich snippets are everywhere – even email!

[Twitter: Miley Ray Cyrus](#)

www.twitter.com/mileycyrus



[mileycyrus](#) Miley Ray Cyrus [Favorite](#) [Retweet](#) [Reply](#)
SMILERS!! WCS is almost at 200 million views!! Shout out to @eos for supporting me and this EPIC video!! pic.twitter.com/gFeh7HvBxV · 6 hours ago



[mileycyrus](#) Miley Ray Cyrus [Favorite](#) [Retweet](#) [Reply](#)
RT @Pharrell "And away we go".. "@MileyCyrus"



Booking.com

Your booking at YOTEL - Times Square

[Modify reservation](#)

[Samsung Galaxy S III Review - Watch CNET's Video Review](#)

reviews.cnet.com/samsung-galaxy-s3-review

★★★★★ Rating: 4/5 · 126 reviews · Published 2012-06-19 · [Specs](#) · [Compare](#)
2012-06-19 · Pumped with high-performing hardware and creative software features, the Samsung Galaxy S3 is an excellent, top-end phone that's neck and neck with the HTC ...

[Flights from Vancouver, Canada \(YVR\) to New York, NY \(all airports\)](#)

www.google.com/flights

[Spam, spam, spam: Twitter's arms race - The Guardian](#)



www.theguardian.com > [News](#) > [Technology](#) > [Twitter](#)

by Charles Arthur - in 5,513 Google+ circles

Apr 7, 2011 - Spam, spam, spam: Twitter's arms race ... Dealing with Twitter

spam occupies seven people full-time, plus the Purchase in stock potentially puts Twitter ahead of Facebook in the ad game.

Wed October 2



h 40m

from \$829

[Images of vladimir putin](#)

bing.com/images



[More Google flight results](#)

[All flights](#)

Air Canada

WestJet

Other airlines

Measurement woes: it's still all about keywords

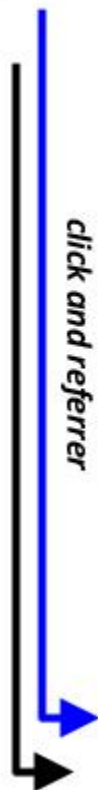
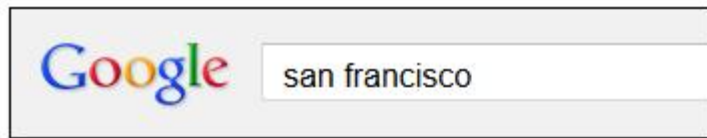
Conventional analytics make it difficult to assess semantic SEO success

Keywords	Position	Previous	Change	Page	Best		
Query	Impressions	Not Changed	Added	Dropped	Ranked	Not Ranked	8
☆ bing video sitemap	50	—	5 —	—	263 ▲ 103	37 ▼ 103	10
☆ paella recipe	30	—	—	—	—	—	25
☆ authentic paella valenciana recipe	<10	—	—	—	—	—	—
☆ paella	<10	—	—	—	—	—	—
☆ sitemap video	<10	—	—	—	—	—	—
☆ bing webmaster tools	700	—	—	—	—	—	—
☆ bing webmaster	90	—	—	—	—	—	—
☆ seo	50	—	—	—	—	—	—
☆ copper prices	2	—	—	—	—	—	—
■ spot gold	—	—	—	—	—	—	—
■ gold price per ounce	—	—	—	—	—	—	—
■ silver spot price	—	—	—	—	—	—	—
First Place	In Top 3	In Top 5	In Top 10				
50 ▲ 30	93 ▲ 55	127 ▲ 59	173 ▲ 73				

Keyword	Visits ?	
	11,304	
% of Total: 50.84% (22,234)		
11. semantic web 101	11	
12. newspaper proposal	10	
13. schema org html meta tags	10	
14. seo markup	10	
15. skeptic or sceptic	10	
16. owl logic trust proof	9	
17. agile seo	8	

Strings and things don't always mix

Visits from semantic search can obscure the original query



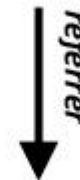
Upcoming events

 18th Annual Ghirardelli... Sep 14	 Method Man & Redman Sep 19	 Kathleen Madigan Sep 14	 Miles for Migraine Sep 14
---	--	---	---

Points of interest

 Golden Gate Bridge	 Alcatraz Island	 Golden Gate Park	 San Francisco Zoo
---	--	---	--

click



Golden Gate Bridge
www.goldengatebridge.org/ ▼
Golden Gate Bridge Highway and Transportation District.
4.6 ★★★★★ 177 Google reviews · [Write a review](#)

 Golden Gate Bridge San Francisco, CA 94129
(415) 921-5858
[Visitors - Tolls & Traffic Operations](#)

Thanks!

Hope you enjoyed the presentation – please say in touch!

Semantic Search Marketing Community on Google+:

Lively discussions! Pie! (Okay - actually no pie)

bit.ly/semsearch

Twitter: [@aaranged](https://twitter.com/aaranged)

Google+: bit.ly/aaranged

Web: seoskeptic.com



see more presentations at:

<http://www.slideshare.net/SearchMarketingExpo>